

Def from last time Let f be a function.

We say that $\lim_{x \rightarrow a} f(x) = l$ if for $\forall \varepsilon > 0, \exists \delta > 0$ s.t.

whenever $0 < |x - a| < \delta$, then

$$|f(x) - l| < \varepsilon$$

Non-example

$$f(x) = \begin{cases} x & \text{if } x \geq 1 \\ -x & \text{if } x < 1 \end{cases}$$

Show that $\lim_{x \rightarrow 1} f(x)$ DNE.

Solution

If $\varepsilon = 3$ is given

(Want to prove $\lim_{x \rightarrow 1} f(x) \neq l$.)

$\Rightarrow$ If $|x - 1| < \delta \Rightarrow |f(x) - 1| < 3 = \varepsilon$.

But if $\varepsilon = 1$ is given, then we can't find a δ .

If $0 < |x - 1| < \delta$ for any $\delta > 0$.

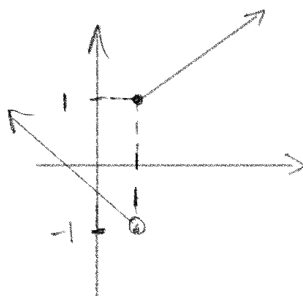
Then $1 - \delta < x - 1 < 1 + \delta$

But $f(1 + \delta) = 1 + \delta$

$$f(1 - \delta) = -(1 - \delta) = -1 + \delta$$

$$\text{So } |f(x) - 1| = \begin{cases} 1 + \delta - 1 = \delta, & \text{if } x > 1. \\ -1 + \delta - 1 = -2 + \delta & \text{if } x < 1. \end{cases}$$

So $|f(x) - 1|$ can take value arbitrarily close to 1 and -1 for whatever δ you find.



Exercise. Show $\lim_{x \rightarrow 1} f(x) \neq l$ for any l .

Pf. Take $\varepsilon = 1$ and suppose $\delta > 0$ exists to satisfy the definition of limit.

look $x = 1 - \varepsilon/2 \in (1 - \delta, 1 + \delta)$.

$$\text{Then } |f(x) - l| = |-1 + \varepsilon/2 - l| = |-2 + \varepsilon/2| = 2 - \varepsilon/2 > 1 = \varepsilon$$

Ex. Show $\lim_{x \rightarrow 5} \frac{1}{x} = \frac{1}{5}$

Pf. let $\varepsilon > 0$. Want to find a δ st.

whenever $0 < |x - 5| < \delta$

We have $|\frac{1}{x} - \frac{1}{5}| < \varepsilon$.

$$\begin{aligned} \text{Notice } |\frac{1}{x} - \frac{1}{5}| &= \left| \frac{5-x}{5x} \right| \\ &= \frac{|x-5|}{|5x|} < \frac{\delta}{|5x|} \end{aligned}$$

Declare that $\delta < 1$.

In this case, $4 < x < 6$

$$\frac{1}{6} < x < \frac{1}{4}$$

$$\text{then } \frac{1}{30} < x < \frac{1}{20}$$

Then we can approximate

$$|f(x) - l| = \left| \frac{1}{x} - \frac{1}{5} \right| < \frac{\delta}{|5x|} < \frac{\delta}{20} < \varepsilon$$

We claim that $\delta = \min\{20\varepsilon, 1\}$.

$$\text{In this case, } |f(x) - l| < \frac{\delta}{20} < \min\{\varepsilon, \frac{1}{20}\} \leq \varepsilon$$



Ex. Show $\lim_{x \rightarrow 0} \sin x = 0$

Solution. Take $\varepsilon > 0$. Want to find $\delta > 0$.

s.t. if $0 < |x - 0| < \delta$, then $|f(x) - 0| < \varepsilon$.

Assume $|\sin x| \leq |x|$, for all $x \in \mathbb{R}$

$$|\sin x - 0| = |\sin x| \leq |x| < \delta = \varepsilon$$

So take $\delta = \varepsilon$

Show $\lim_{x \rightarrow a} \sin x = \sin a$

Let $\varepsilon > 0$, want to find $\delta > 0$. s.t.

whenever $0 < |x - a| < \delta$,

we have $|\sin x - \sin a| < \varepsilon$

$$\begin{aligned} & |\sin x - \sin a| \\ = & \left| 2 \sin\left(\frac{x-a}{2}\right) \cos\left(\frac{x+a}{2}\right) \right| \\ \leq & \left| 2 \left| \frac{x-a}{2} \right| \cos\left(\frac{x+a}{2}\right) \right| \\ < & \delta \left| \cos\left(\frac{x+a}{2}\right) \right| \\ \leq & \delta < \varepsilon \end{aligned}$$

Take $\delta = \frac{\varepsilon}{2}$

Then $|f(x) - \sin a| = |\sin x - \sin a| < \delta = \frac{\varepsilon}{2}$.

